

Practical No.: 5

Course- B. Tech.	Type- Core
Course Code- CSE25402P	Course Name- Database Management Systems
Year- 2026	Semester- Even
Date- 09/01/2026	Batch- 2024-2028

Consider the Company database given below and execute the following queries:

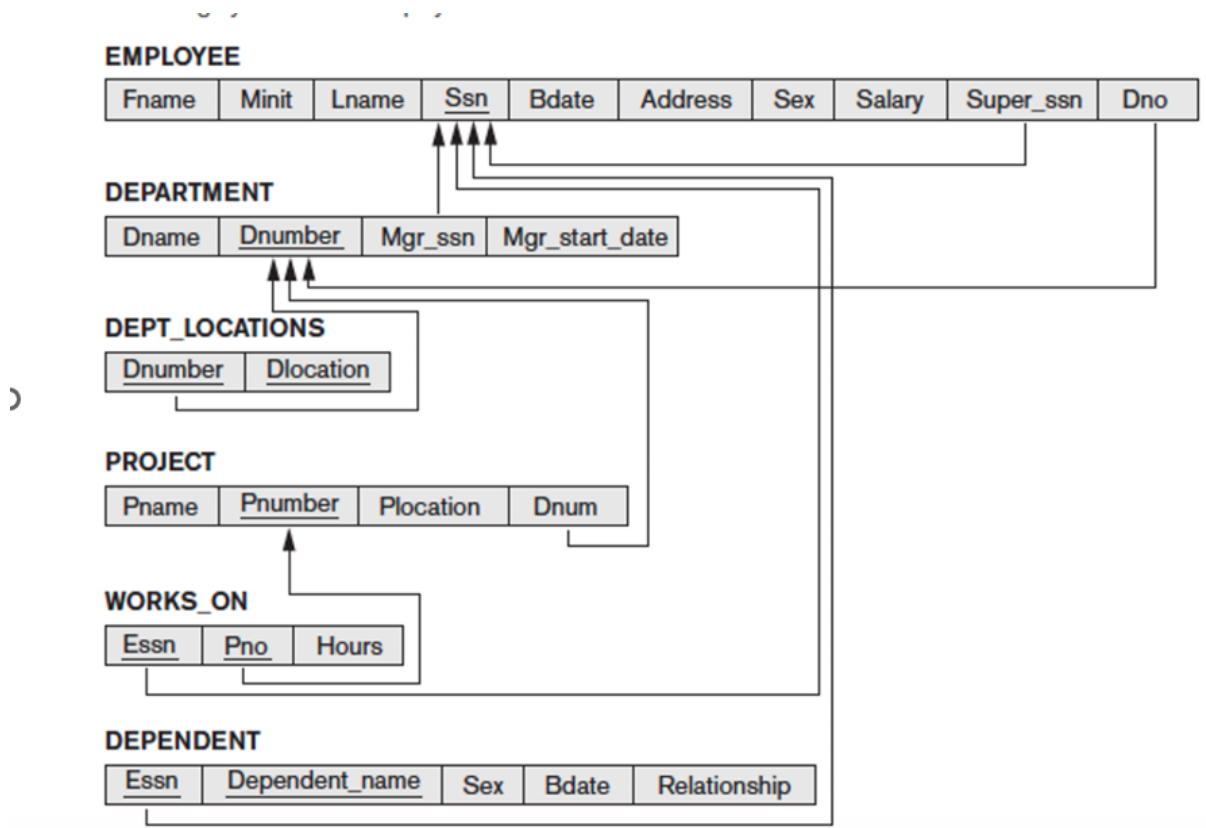


Figure 1. Relational Schema for Company Database

Create Schema using the following using DDL commands:

CREATE TABLE EMPLOYEE

(Fname	VARCHAR(15)	NOT NULL,
Minit	CHAR,	
Lname	VARCHAR(15)	NOT NULL,
Ssn	CHAR(9)	NOT NULL,
Bdate	DATE,	
Address	VARCHAR(30),	
Sex	CHAR,	
Salary	DECIMAL(10,2),	
Super_ssn	CHAR(9),	
Dno	INT	NOT NULL,

PRIMARY KEY (Ssn),

CREATE TABLE DEPARTMENT

(Dname	VARCHAR(15)	NOT NULL,
Dnumber	INT	NOT NULL,
Mgr_ssn	CHAR(9)	NOT NULL,
Mgr_start_date	DATE,	

PRIMARY KEY (Dnumber),

UNIQUE (Dname),

FOREIGN KEY (Mgr_ssn) **REFERENCES** EMPLOYEE(Ssn));

CREATE TABLE DEPT_LOCATIONS

(Dnumber	INT	NOT NULL,
Dlocation	VARCHAR(15)	NOT NULL,

PRIMARY KEY (Dnumber, Dlocation),

FOREIGN KEY (Dnumber) **REFERENCES** DEPARTMENT(Dnumber));

CREATE TABLE PROJECT

(Pname	VARCHAR(15)	NOT NULL,
Pnumber	INT	NOT NULL,
Plocation	VARCHAR(15),	
Dnum	INT	NOT NULL,

PRIMARY KEY (Pnumber),

UNIQUE (Pname),

FOREIGN KEY (Dnum) **REFERENCES** DEPARTMENT(Dnumber));

CREATE TABLE WORKS_ON

(Essn	CHAR(9)	NOT NULL,
Pno	INT	NOT NULL,
Hours	DECIMAL(3,1)	NOT NULL,

PRIMARY KEY (Essn, Pno),

FOREIGN KEY (Essn) **REFERENCES** EMPLOYEE(Ssn),

FOREIGN KEY (Pno) **REFERENCES** PROJECT(Pnumber));

CREATE TABLE DEPENDENT

(Essn	CHAR(9)	NOT NULL,
Dependent_name	VARCHAR(15)	NOT NULL,
Sex	CHAR,	
Bdate	DATE,	
Relationship	VARCHAR(8),	

PRIMARY KEY (Essn, Dependent_name),

FOREIGN KEY (Essn) **REFERENCES** EMPLOYEE(Ssn));

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Figure 2. Data to be populated in Relational Schema.

Execute the following queries:

A. DML commands:

1. Refer to the schema given above create all tables with primary, foreign key, not null, default and Check constraints and populate all the tables as shown in Figure 2
2. Add column salary to table dept and drop the column mgrs date.
3. Insert <Robert', 'F', 'Scott', '943775543', '1952-06-21', '2365 Newcastle Rd, Bellaire, TX', M, 58000, '888665555',1> into EMPLOYEE.
4. Insert <'ProductA', 4, 'Bellaire', 2> into PROJECT.
5. Insert <'Production', 4, '943775543', '1998-10-01'> into DEPARTMENT.
6. Insert <'677678989', null, '40.0'> into WORKS_ON.

7. Insert <'453453453', 'John', M, '1970-12-12', 'SPOUSE'> into DEPENDENT.
8. Delete the WORKS_ON tuples with ESSN = '333445555'.
9. Delete the EMPLOYEE tuple with SSN = '987654321'.
10. Delete the PROJECT tuple with PNAME = 'ProductX'.
11. Modify the MGRSSN and MGRSTARTDATE of the DEPARTMENT tuple with DNUMBER = 5 to '123456789' and '1999-10-01', respectively.
12. Modify the SUPERSSN attribute of the EMPLOYEE tuple with SSN = '999887777' to '943775543'.
13. Modify the HOURS attribute of the WORKS_ON tuple with ESSN = '999887777' and PNO = 10 to '5.0'.
14. List the name of employees whose salary is between 30000 and 50000.
15. List the name of all employees who work in department 5.
16. List the name of all employees who work in dept no.2 and whose name starts with 'A'.
17. List the name of all Female employees.
18. List the total salary of all employees who lives in Noida.
19. Find the project name and project no. where Mgrsdate is on or before June 11 2014.
20. Find fname, lname, ssn, bdate of all the employees and sort the results according to bdate.
21. Find the average of salary for all the employees where bdate is between jan 1 1990 to jan 1 2000.